Biology practice paper: mark scheme

For Dad. Keep this away from Dougie until he has finished the paper.

Total: 30 marks **Written in the style of AQA mark schemes**

How to mark

- A slash (/) separates answers that are each worth the mark. Words in brackets are not needed for the mark.
- Give the mark for any answer that means the same as the one listed, even in different words.
- In calculations, give full marks for the correct answer with no working, and the working mark if the answer is wrong but the method is right.
- Or photograph each page of his answers and send them to Claude to mark against this scheme with feedback.

Question 1: Cells (5 marks)

Part	Answer	Mark
1.1	Chloroplast Only one box ticked. Two or more ticked: no mark.	**1**
1.2	6 ÷ 0.06 (1) = (×) 100 (1) Correct answer with no working: 2 marks. Do not accept 0.01.	**2**
1.3	Any two from: the cell grows / gets bigger (1) the DNA is copied / the chromosomes are copied (1) the number of sub-cellular structures increases, such as ribosomes or mitochondria (1) Do not accept "the nucleus divides": that is mitosis itself.	**2**

Question 2: Osmosis practical (7 marks)

Part	Answer	Mark
2.1	(2.80 - 2.50) = 0.30 and 0.30 ÷ 2.50 × 100 (1) = (+) 12 (%) (1) Correct answer with no working: 2 marks. Accept +12.0. Do not accept -12.	**2**
2.2	0.2 and 0.4 (mol/dm³) Both needed, in either order.	**1**
2.3	Water moved out of the potato (cells) by osmosis (1) because the solution was more concentrated than the inside of the potato cells / the potato cells were more dilute (1)	**3**

	water moves from a dilute solution to a more concentrated solution through a partially permeable membrane / through the cell membrane (1) Do not accept "sugar moved in" or "the potato lost sugar". Ignore references to water potential if used correctly.	
2.4	The cylinders had different starting masses / so the results can be compared (fairly) Do not accept "it's more accurate" on its own.	**1**

Question 3: Enzymes and food tests (6 marks)

Part	Answer	Mark
3.1	Lipase (1) fatty acids and glycerol (1) Both products needed for the second mark. Do not accept "fats" or "amino acids".	**2**
3.2	The active site changes shape / the enzyme is denatured (1) so the substrate no longer fits (into the active site) (1) Do not accept "the enzyme is killed" or "the enzyme dies": no marks for that idea.	**2**
3.3	Biuret (reagent / solution) (1) purple / lilac / mauve (1) Ignore the colour before the test (blue). Do not accept "pink" on its own.	**2**

Question 4: Coronary heart disease (2 marks)

Part	Answer	Mark
4.1	It keeps the coronary artery open / widens the (narrowed) coronary artery (1) so blood (carrying oxygen) can flow to the heart muscle (1) "Artery" without "coronary" can get the first mark if the answer is clearly about the heart. Do not accept "unblocks a vein".	**2**

Question 5: Infection and response (4 marks)

Part	Answer	Mark
5.1	Antibiotics only kill bacteria / antibiotics don't kill viruses / viruses live inside cells (where antibiotics can't reach them) Do not accept "the virus is resistant".	**1**
5.2	A small amount of dead / inactive (form of the) measles virus is injected (1) white blood cells make antibodies (against it) (1) if the real virus gets in later, white blood cells make the right antibodies quickly (and in large numbers), so the virus is destroyed before the person becomes ill (1) Accept "the body remembers" for the third mark only if it's linked to antibodies being made faster. Do not accept "the vaccine contains antibodies".	**3**

Question 6: Photosynthesis practical (6 marks)

Mark this as a whole, in levels, like the real 6-mark questions. First choose the level that best describes the answer, then the mark within it.

Level	What the answer does	Marks
Level 3	A clear method in a logical order that would work. It says how light intensity is changed, how the rate of photosynthesis is measured, and at least two things kept the same, with a repeat or mean.	**5 to 6**
Level 2	A method that would mostly work, but with a step missing or unclear. It changes light intensity and measures something, and mentions at least one thing kept the same.	**3 to 4**
Level 1	Some relevant steps, but no clear order, or it doesn't say how the rate is measured.	**1 to 2**
No marks	Nothing relevant.	**0**

Points that could be in the answer

- Put the pondweed in a boiling tube or beaker of water (you can add sodium hydrogencarbonate to give it plenty of carbon dioxide).
- Put a lamp at a set distance from the pondweed, such as 10 cm, measured with a ruler.
- Leave it for a few minutes to settle before starting.
- Count the bubbles of oxygen given off in a set time, such as one minute, or collect the gas and measure its volume.
- Move the lamp to different distances, such as 20, 30, 40 and 50 cm, and repeat the count at each one.
- Repeat each distance and work out a mean.
- Keep the same: the temperature (put a beaker of water between the lamp and the plant, or use an LED bulb), the same pondweed, the same amount of carbon dioxide, the same time for each count, and no other light (darkened room).

> **Totals:** Q1 5, Q2 7, Q3 6, Q4 2, Q5 4, Q6 6. Total 30. As a rough guide, 22 or more is a strong result, 15 to 21 is solid with gaps to fix, and under 15 means going back over these topics in the notebook first. These bands are a guide for this practice paper only, not AQA grade boundaries.